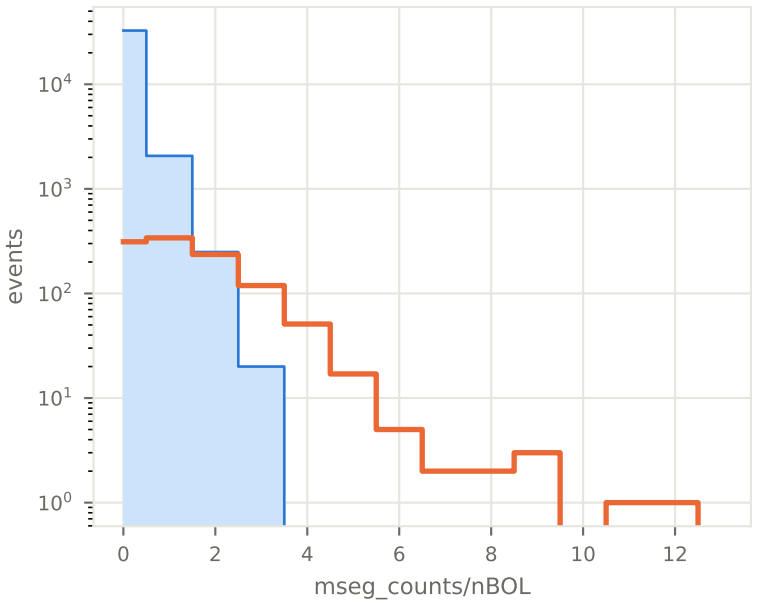
mseg_counts/nBOL

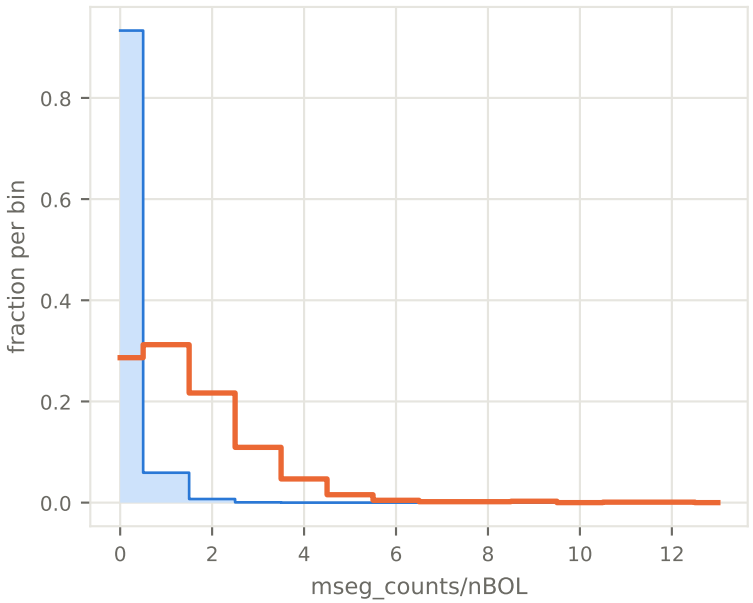
BARREL vs ENDCAP · shape · rank 40 / 98 by |D_barrel - D_endcap| · neither

shape D(barrel vs endcap): background 0.647 signal 0.626 separation D: endcap 0.057 barrel 0.132

1a · background (data) · raw

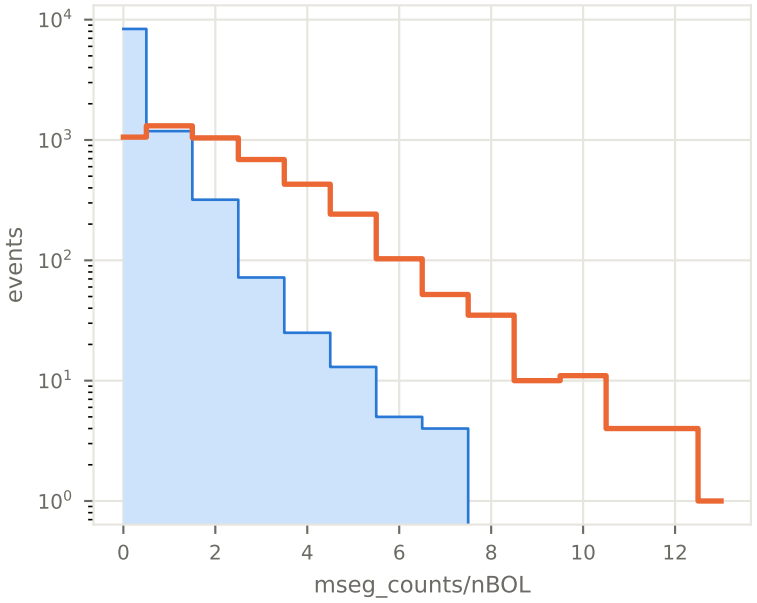


1b · background (data) · normalized

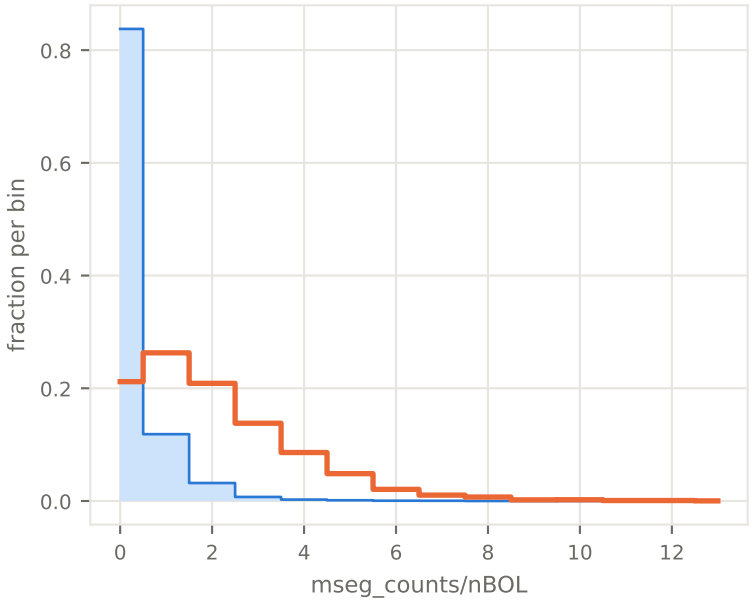


1 · background (data)
shape D = 0.647
endcap n 34,970
barrel n 1,089

2a · combined signal (MC, pooled) · raw



2b · combined signal (MC, pooled) · normalized



2 · combined signal (MC, pooled)
shape D = 0.626
endcap n 9,996
barrel n 4,987

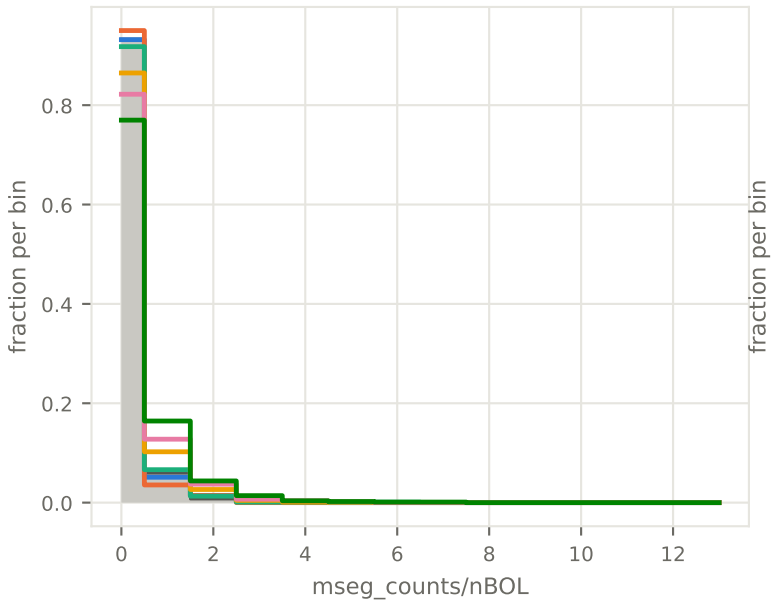
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

mseg_counts/nBOL

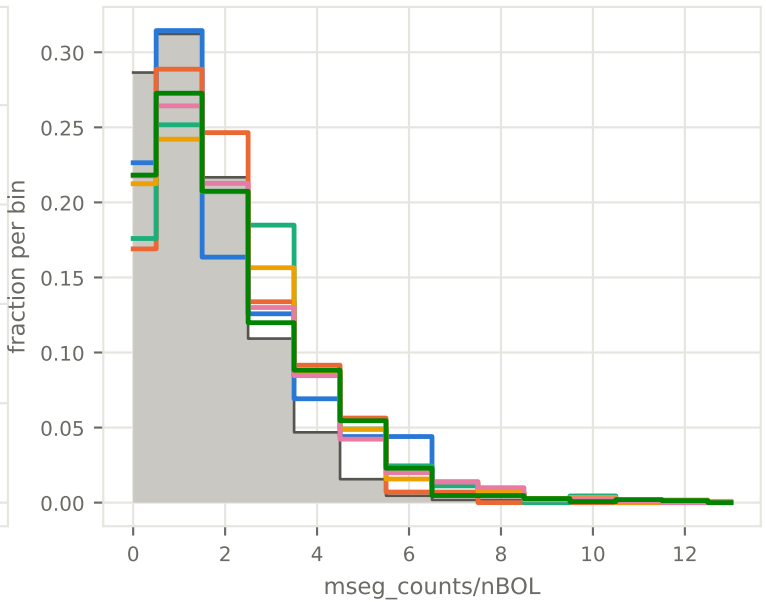
BARREL vs ENDCAP · separation side by side · rank 40 / 98 · neither

D_equal endcap 0.057   barrel 0.132   difference (barrel - endcap) +0.075   (normalized only: raw counts differ ~30x between regions)

1a · ENDCAP · each mass point vs background

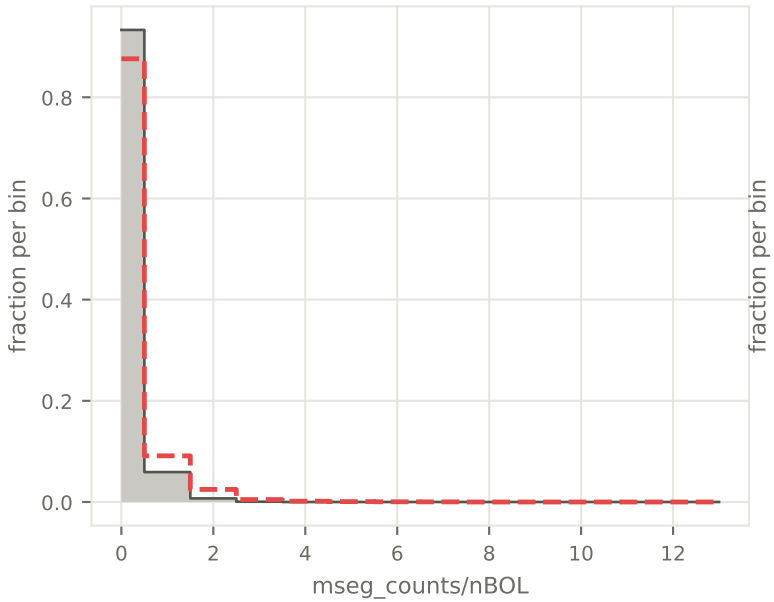


1b · BARREL · each mass point vs background

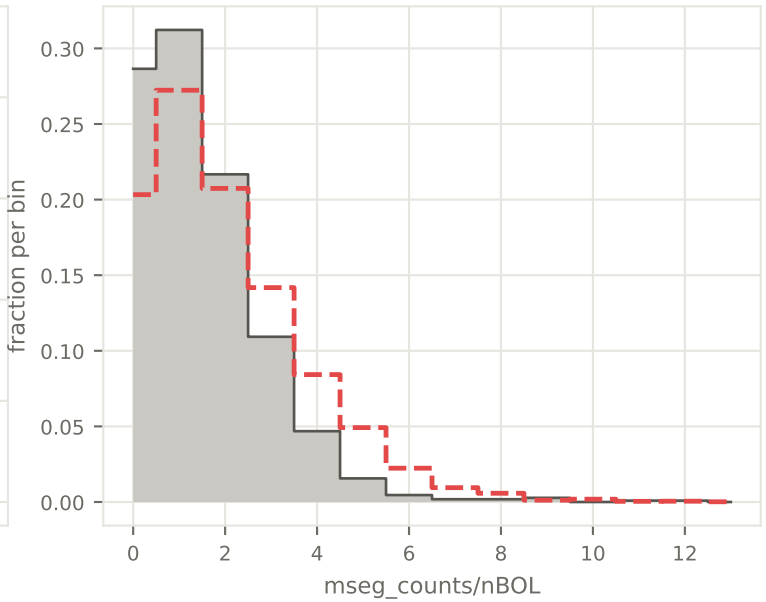


1 · each mass point
D: endcap | barrel
background n 34,970 | 1,089
mH125_mS15 D 0.009 | 0.111
mH125_mS55 D 0.017 | 0.141
mH200_mS50 D 0.015 | 0.181
mH400_mS100 D 0.068 | 0.154
mH600_mS150 D 0.111 | 0.121
mH1000_mS275 D 0.163 | 0.117

2a · ENDCAP · combined signal vs background



2b · BARREL · combined signal vs background



2 · combined signal
D: endcap | barrel
background n 34,970 | 1,089
combined, equal weight D 0.057 | 0.132

a = endcap b = barrel · normalized · edge bins include under/overflow · NaN and -999 removed